

PHISHING DETECTION & AWARENESS REPORT

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PROJECT OVERVIEW



This project focuses on identifying common phishing indicators in email messages and creating practical security awareness guidance for users. The assessment covers sender information, urgency, threatening language, suspicious links, information requests, email authentication results, and social-engineering techniques.

Three educational email samples were reviewed:

- Amazon-style account restriction phishing email
- Trading/investment account verification phishing email
- Legitimate account statement notification

A synthetic email header was also analyzed to demonstrate SPF, DKIM, DMARC, sender-domain, and Reply-To analysis.

All phishing emails and headers used in this project are synthetic educational samples. No real phishing email was sent, and no real account or credential was targeted.

Objectives



The objectives of this project are:

- Identify common phishing indicators.
- Analyze suspicious sender domains and links safely.
- Review basic email header information.
- Understand SPF, DKIM, and DMARC authentication results.
- Classify emails according to their risk.
- Explain phishing attacks in simple terms.
- Develop practical phishing prevention guidelines.
- Create employee Do's and Don'ts for phishing awareness.

TOOLS AND RESOURCES



Tools	Techniques
Web browser	Sender-address inspection
Text editor	Domain analysis
Google Message Header Analyzer	Email-header analysis
GitHub	SPF/DKIM/DMARC review
	Risk classification
	Security awareness documentation
	Link inspection without opening suspicious links

PHISHING EMAIL ANALYSIS



Sample 01 – Amazon-Style Account Alert

Classification: PHISHING – HIGH RISK

Indicators identified:

- Suspicious sender domain
- Urgent 24-hour deadline
- Account-suspension threat
- Generic greeting
- Suspicious verification link
- Request for account information

Assessment:

The email uses urgency and fear of account suspension to encourage the recipient to click a verification link.

Recommended action: Do not click the link or provide account information. Access the service through its official application or website.

PHISHING EMAIL ANALYSIS



Sample 02 – Trading / Investment Account Alert

Classification: PHISHING – HIGH RISK

Indicators identified:

- Suspicious trading-related domain
- 12-hour deadline
- Financial-account targeting
- Account-restriction threat
- Suspicious verification link
- Social-engineering pressure

Assessment:

The email creates fear about account access and funds, then pressures the recipient to verify the account through a suspicious link.

Recommended action: Do not click the link or enter credentials. Use the official trading application or website.

PHISHING EMAIL ANALYSIS



Sample 03 – Account Statement Notification

Classification: SAFE / LOW RISK

Observations:

- Expected sender-domain format
- Personalized greeting
- No threatening deadline
- No suspicious login link
- No request for passwords or OTPs
- Directs the user to the official application or website

Assessment:

The email does not contain the major phishing indicators observed in the previous samples.

EMAIL HEADER ANALYSIS



Classification: PHISHING – HIGH RISK

A synthetic training header was created to demonstrate email authentication analysis.

Key findings:

- SPF: FAIL – Sender authentication failed.
- DKIM: NONE – No DKIM signature was present.
- DMARC: FAIL – The message failed DMARC authentication.
- Sender Domain: Suspicious domain attempting to imitate a trusted organization.
- Reply-To: Points to the suspicious sender domain.

Assessment:

The combination of authentication failures and a suspicious sender domain increases the likelihood that the message is fraudulent.

Email authentication results should be considered together with other phishing indicators.

RISK CLASSIFICATION



Sample	Classification	Risk
Amazon-style account alert	Phishing	High
Trading account alert	Phishing	High
Account statement notification	Safe	Low
Synthetic header	Phishing	High

COMMON PHISHING INDICATORS



The analysis identified the following common indicators:

- Suspicious sender domains
- Display-name impersonation
- Urgent deadlines
- Threatening language
- Generic greetings
- Suspicious links
- Requests for sensitive information
- Financial targeting
- Suspicious Reply-To addresses
- SPF/DKIM/DMARC failures
- Social engineering

PHISHING AWARENESS – DO'S & DON'TS



DO

- Check the complete sender address.
- Inspect links before clicking.
- Verify suspicious requests through official channels.
- Report suspicious emails to IT/security teams.
- Use multi-factor authentication.
- Verify unexpected payment or account-change requests.

DON'T

- Don't click suspicious links.
- Don't open unexpected attachments.
- Don't share passwords, OTPs, PINs, or banking information.
- Don't reply to suspicious emails.
- Don't trust the displayed sender name alone.
- Don't bypass security warnings.

IF YOU CLICKED A SUSPICIOUS LINK

1. Stop entering information.
2. Close the webpage.
3. Report the incident.
4. If credentials were entered, change the password through the legitimate website.
5. Change reused passwords elsewhere.
6. Enable MFA where available.
7. Follow organizational incident-response procedures.

CONCLUSION



This project demonstrated practical phishing detection through sender analysis, link inspection, social-engineering identification, risk classification, and email-header analysis.

The exercise also produced practical awareness guidelines to help users recognize and safely respond to phishing emails.

STOP → CHECK → VERIFY → REPORT

DISCLAIMER

All phishing emails and email headers used in this project are synthetic educational samples created for cybersecurity training.

No real phishing campaign was conducted, no credentials were collected, and no real user account was targeted.

